

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	K Abishek	Reg. No.:	192521120
Section / Batch:	IT	Date:	16/07/26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Smart Classroom Attendance System

Q2. Remote Sensing Image Processor

Code of Conduct

I K Abishek/192521120 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K Abishek

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Smart Classroom Attendance System

Problem Statement

A smart classroom attendance system scans each student ID exactly once while marking attendance. Analyze the time complexity and space complexity, explain how the number of operations and memory usage grow with the input size, and determine the asymptotic complexities.

Time Complexity

Time complexity measures how the execution time of an algorithm grows as the input size (**n**) increases. It is represented using **Big-O notation**.

In the attendance system:

- Each student ID is scanned exactly once.
- Each scan and attendance update takes constant time **O(1)**.
- For **n** students, the scan is repeated **n** times.

Therefore, the total number of operations is:

$$T(n) = n$$

Hence,

Time Complexity = O(n)

This is called **Linear Time Complexity**, since the execution time increases linearly with the number of students.

Growth of Number of Operations

Number of Students (n) Operations

10	10
100	100
1000	1000
n	n

The number of operations grows directly in proportion to the input size.

Space Complexity

Space complexity measures the amount of memory used during execution.

The algorithm only requires:

- Current student ID
- Loop counter

- Attendance status

No additional memory proportional to **n** is allocated.

Therefore,

Auxiliary Space = O(1)

Hence,

Space Complexity = O(1)

The memory usage remains constant even if the number of students increases.

Asymptotic Complexity

- **Time Complexity: O(n)**
- **Space Complexity: O(1)**

Conclusion

Since every student is processed exactly once, the execution time grows linearly with the number of students. The algorithm uses only a constant amount of extra memory, making it efficient for handling large classroom attendance.

Remote Sensing Image Processor

Problem Statement

A satellite image is recursively divided into two equal parts, with additional linear work performed at each step.

The recurrence relation is:

$$T(n) = 2T(n/2) + n$$

Determine the time complexity using the Master Theorem and analyze the space complexity.

Master Theorem

The Master Theorem solves recurrence relations of the form:

$$T(n) = aT(n/b) + f(n)$$

Where:

- **a** = Number of recursive subproblems
- **b** = Factor by which the input size is reduced
- **f(n)** = Work done outside the recursive calls

Identify the Parameters

From

$$T(n) = 2T(n/2) + n$$

we get:

- $a = 2$
- $b = 2$
- $f(n)=n$

Now calculate

$$n^{\log_b a} = n^{\log_2 2} = n$$

Since

$$f(n) = \Theta(n^{\log_b a})$$

the recurrence satisfies **Case 2** of the Master Theorem.

Master Theorem Cases

Case 1

If

$$f(n) = O(n^{\log_b a - \epsilon})$$

Then

$$T(n) = \Theta(n^{\log_b a})$$

Case 2

If

$$f(n) = \Theta(n^{\log_b a} \log^k n)$$

Then

$$T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$$

This is the case applicable to the given recurrence.

Case 3

If

$$f(n) = \Omega(n^{\log_b a + \epsilon})$$

and the regularity condition holds,

$$T(n) = \Theta(f(n))$$

Time Complexity

Applying Case 2,

$$T(n) = \Theta(n \log n)$$

Therefore,

Time Complexity = $O(n \log n)$

At every level of recursion, the total work performed is **$O(n)$** , and there are **$\log n$** recursion levels. Hence, the total running time becomes **$O(n \log n)$** .

Space Complexity

The recursive function stores one activation record for each recursive call.

Since the input size is halved at every recursive step,

$$n, \frac{n}{2}, \frac{n}{4}, \frac{n}{8}, \dots$$

the recursion depth is

$$\log_2 n$$

Each recursive call requires constant extra memory.

Therefore,

Space Complexity = $O(\log n)$

Asymptotic Complexity

- **Time Complexity: $O(n \log n)$**
- **Space Complexity: $O(\log n)$**

Conclusion

The Remote Sensing Image Processor follows the **divide-and-conquer** strategy by recursively dividing the image into two equal parts. Using the Master Theorem, the recurrence relation **$T(n)=2T(n/2)+n$** is solved as **$O(n \log n)$** . Since the recursion depth is **$\log n$** , the auxiliary memory required is **$O(\log n)$** .

This version is focused on exactly what the questions ask, without unnecessary theory, and is appropriate for a university assessment.